Cherry Blossom Run, 2017

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Task 1: As always, we start by loading some R packages. Load `tidyverse` `ggdist` and `ggridges` packages below. These will help us with graphing our data.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggdist)
library(ggridges)
```

```
##
## Attaching package: 'ggridges'
##
## The following objects are masked from 'package:ggdist':
##
##     scale_point_color_continuous, scale_point_color_discrete,
##     scale_point_colour_continuous, scale_point_colour_discrete,
##     scale_point_fill_continuous, scale_point_fill_discrete,
##     scale_point_size_continuous
```

The data

Task 2: We will get the data for this activity directly from our textbook's web site. Fill in the code below to load the data. We will all give the dataframe the name run17. We will load the data from the URL https://www.openintro.org/data/csv/run17.csv instead of pointing R to a file in our project.

```
run17<- read.csv("https://www.openintro.org/data/csv/run17.csv")
```

Task 3: Next, run this piece of code to drop the missing values of `pace_sec` from our dataset. This will make performing calculations easier.

```r
run17<-run17 %>%
  drop_na(pace_sec)
```

You thought about this data set in Quiz 1. It has information about the 19,961 runners in the 2017 Cherry Blossom Run.

The full codebook is at https://www.openintro.org/data/index.php?data=run17.

Task 4: Do the usual visual exploration - read the codebook, examine the dataset here in R somehow or another (click on the run17 dataframe that popped up in your environment to the right, or use glimpse or head to view the dataset.)

```r
summary(run17)
```

```
##       bib            name               sex                 age
##  Min.   :    3   Length:19958       Length:19958       Min.   : 6.00
##  1st Qu.: 6077   Class :character   Class :character   1st Qu.:29.00
##  Median :12128   Mode  :character   Mode  :character   Median :35.00
##  Mean   :12299                                         Mean   :37.13
##  3rd Qu.:18320                                         3rd Qu.:44.00
##  Max.   :25497                                         Max.   :92.00
##                                                        NA's   :1
##      city              net_sec       clock_sec       pace_sec
##  Length:19958       Min.   :1078   Min.   :1078   Min.   : 279.0
##  Class :character   1st Qu.:4767   1st Qu.:5048   1st Qu.: 524.0
##  Mode  :character   Median :5658   Median :6333   Median : 596.0
##                     Mean   :5428   Mean   :6083   Mean   : 608.7
##                     3rd Qu.:6433   3rd Qu.:7439   3rd Qu.: 674.0
##                     Max.   :8398   Max.   :9896   Max.   :1759.0
##
##     event
##  Length:19958
##  Class :character
##  Mode  :character
##
##
##
##
```

```r
head(run17, 10)
```

```
##    bib          name sex age                 city net_sec clock_sec pace_sec
## 1    6      Hiwot G.   F  21             Ethiopia    3217      3217      321
## 2   22       Buze D.   F  22             Ethiopia    3232      3232      323
## 3   16     Gladys K.   F  31                Kenya    3276      3276      327
## 4    4     Mamitu D.   F  33             Ethiopia    3285      3285      328
## 5   20   Karolina N.   F  35               Poland    3288      3288      328
## 6    8 Firehiwot D.   F  33             Ethiopia    3316      3316      331
## 7   18       Tara W.   F  27         Portland, OR    3334      3334      333
## 8   24      Nancy N.   F  21                Kenya    3357      3357      335
## 9   44     Hannah D.   F  26 Saratoga Springs, NY    3359      3359      335
## 10  30    Susanna S.   F  26           Reston, VA    3372      3372      337
##      event
## 1  10 Mile
## 2  10 Mile
```

```
## 3  10 Mile
## 4  10 Mile
## 5  10 Mile
## 6  10 Mile
## 7  10 Mile
## 8  10 Mile
## 9  10 Mile
## 10 10 Mile
```

```r
glimpse(run17)
```

```
## Rows: 19,958
## Columns: 9
## $ bib       <int> 6, 22, 16, 4, 20, 8, 18, 24, 44, 30, 38, 23051, 34, 640, 150~
## $ name      <chr> "Hiwot G.", "Buze D.", "Gladys K.", "Mamitu D.", "Karolina N~
## $ sex       <chr> "F", "F", "F", "F", "F", "F", "F", "F", "F", "F", "F", "F", ~
## $ age       <int> 21, 22, 31, 33, 35, 33, 27, 21, 26, 26, 25, 28, 26, 22, 27, ~
## $ city      <chr> "Ethiopia", "Ethiopia", "Kenya", "Ethiopia", "Poland", "Ethi~
## $ net_sec   <int> 3217, 3232, 3276, 3285, 3288, 3316, 3334, 3357, 3359, 3372, ~
## $ clock_sec <int> 3217, 3232, 3276, 3285, 3288, 3316, 3334, 3357, 3359, 3372, ~
## $ pace_sec  <int> 321, 323, 327, 328, 328, 331, 333, 335, 335, 337, 344, 345, ~
## $ event     <chr> "10 Mile", "10 Mile", "10 Mile", "10 Mile", "10 Mile", "10 M~
```

```r
run17 %>%
  group_by(event) %>%
  summarize(count = n())
```

```
## # A tibble: 2 x 2
##   event   count
##   <chr>   <int>
## 1 10 Mile 17442
## 2 5K       2516
```

Task 5: Would you consider this data set to be data from a sample or a population? Write your answer in the space below.

The dataframe represents a sample, not a population. while the dataframe is large having a total of 19,961 entries, it does not represent the entire population of cherry blossom runners across different years or all possible runners in general.

Here are some research questions we will consider.

1. What is the average pace-time among all runners in the Cherry Blossom Run?
2. What is the average pace-time by event?

Notice that the first question is a question about a SINGLE numerical variable, the pace-time of the runners. The second question is about ONE numerical variable (pace-time of the runners) and ONE categorical variable (which event the runner ran).

Task 6: Look up the codebook and copy down the description of the variable pace_sec below.

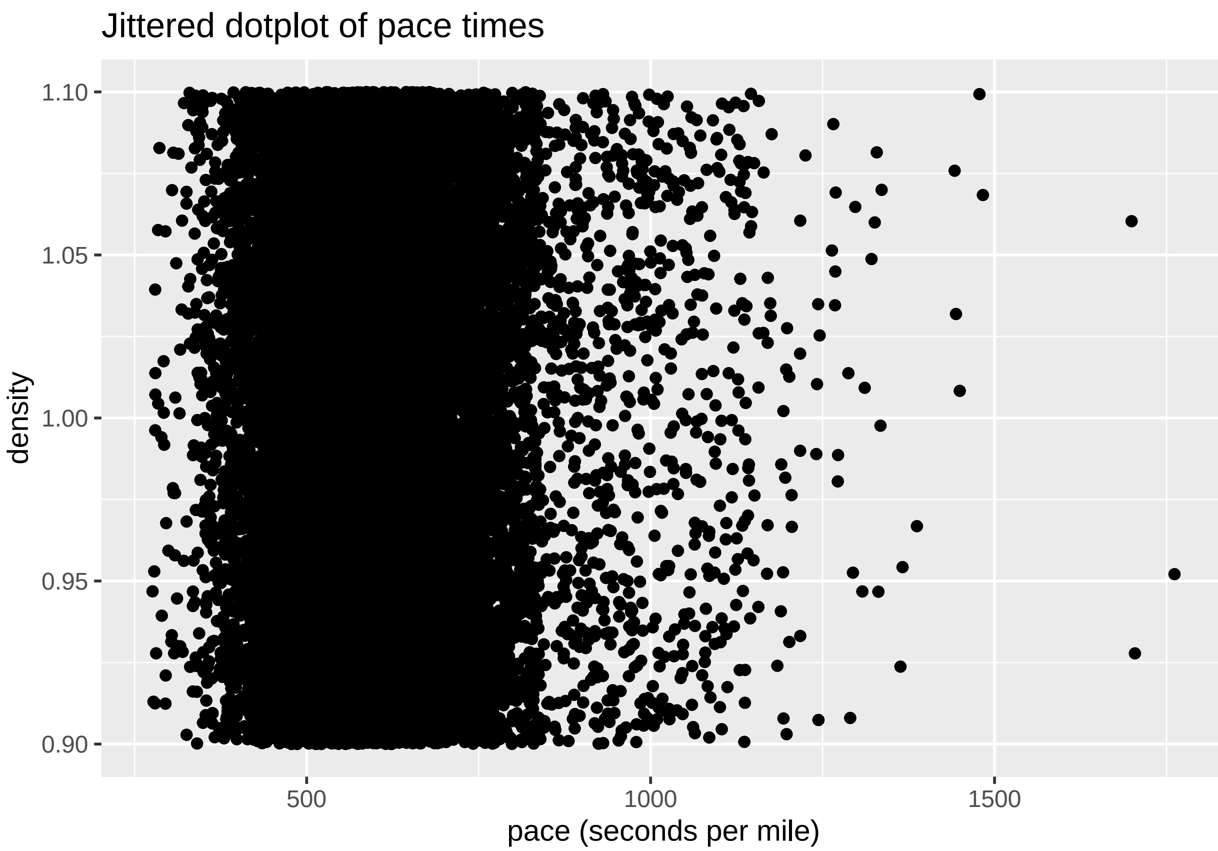
Average time per mile, in seconds.

Task 7: Have R create several graphs displaying the distribution of pace times for all runners. Again, see our class R code cheat sheet to find the code you need.

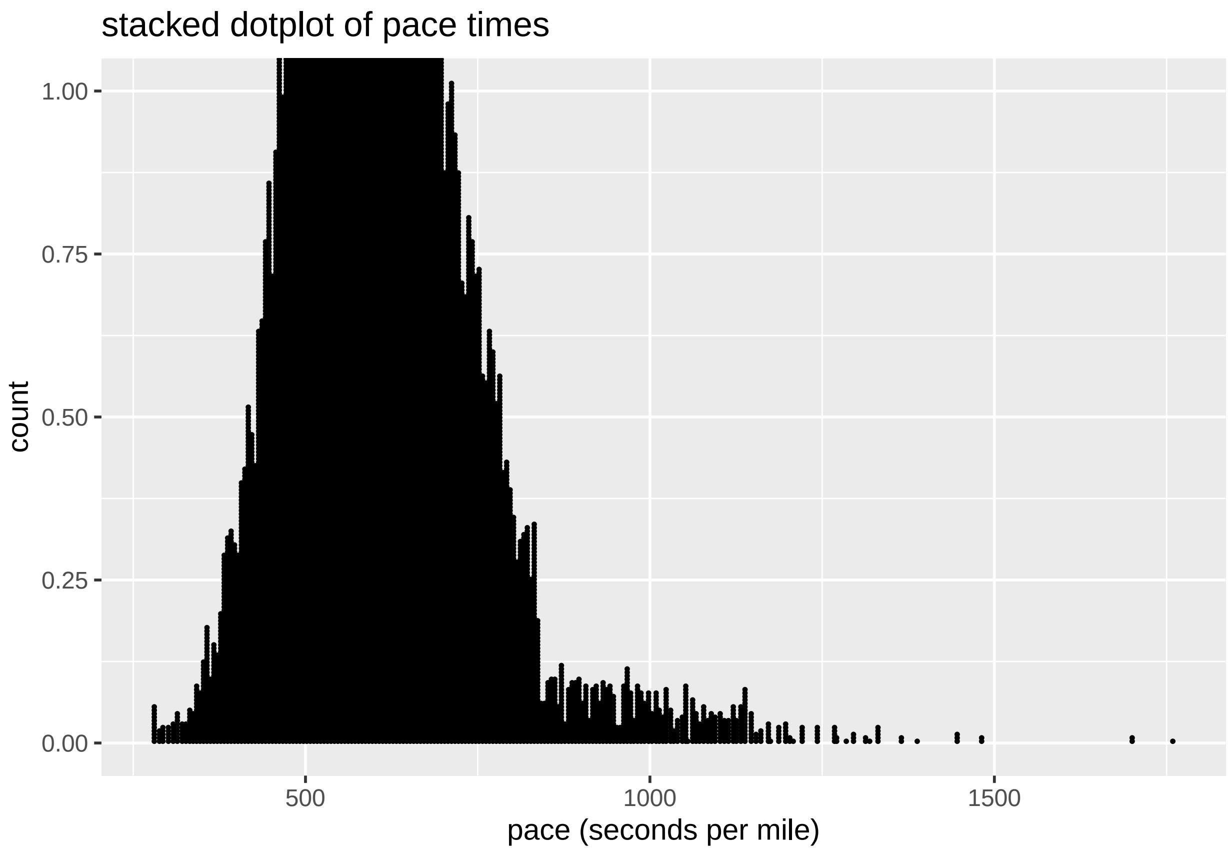
• A jittered dotplot (adjust the height and width) • A stacked dotplot • A histogram (play with the binwidth to find one that looks good) • A frequency polygon.

(See the R Code Cheat Sheet under the R Help Documents tab in Blackboard.)

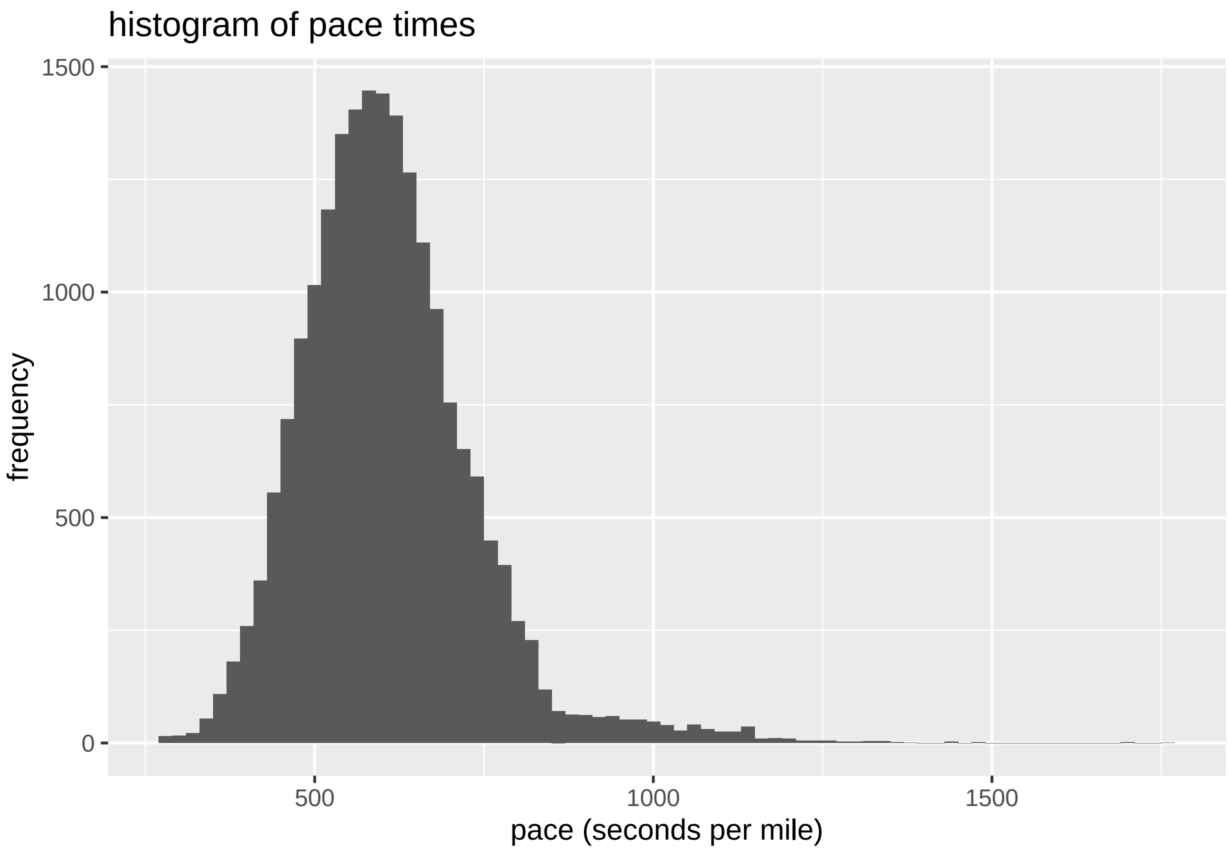
```
# jitter
ggplot(data = run17, aes(x = pace_sec, y = 1)) +
  geom_jitter(width = 5, height = 0.1) +
  labs(title = "Jittered dotplot of pace times", x = "pace (seconds per mile)", y = "density")
```




```
# stacked Dotplot (using basic stacked dot plot data is very close and difficult to see changed "stackd
ggplot(data = run17, aes(x = pace_sec)) +
  geom_dotplot(binwidth = 5, stackdir = "up") +
  labs(title = "stacked dotplot of pace times", x = "pace (seconds per mile)", y = "count")
```

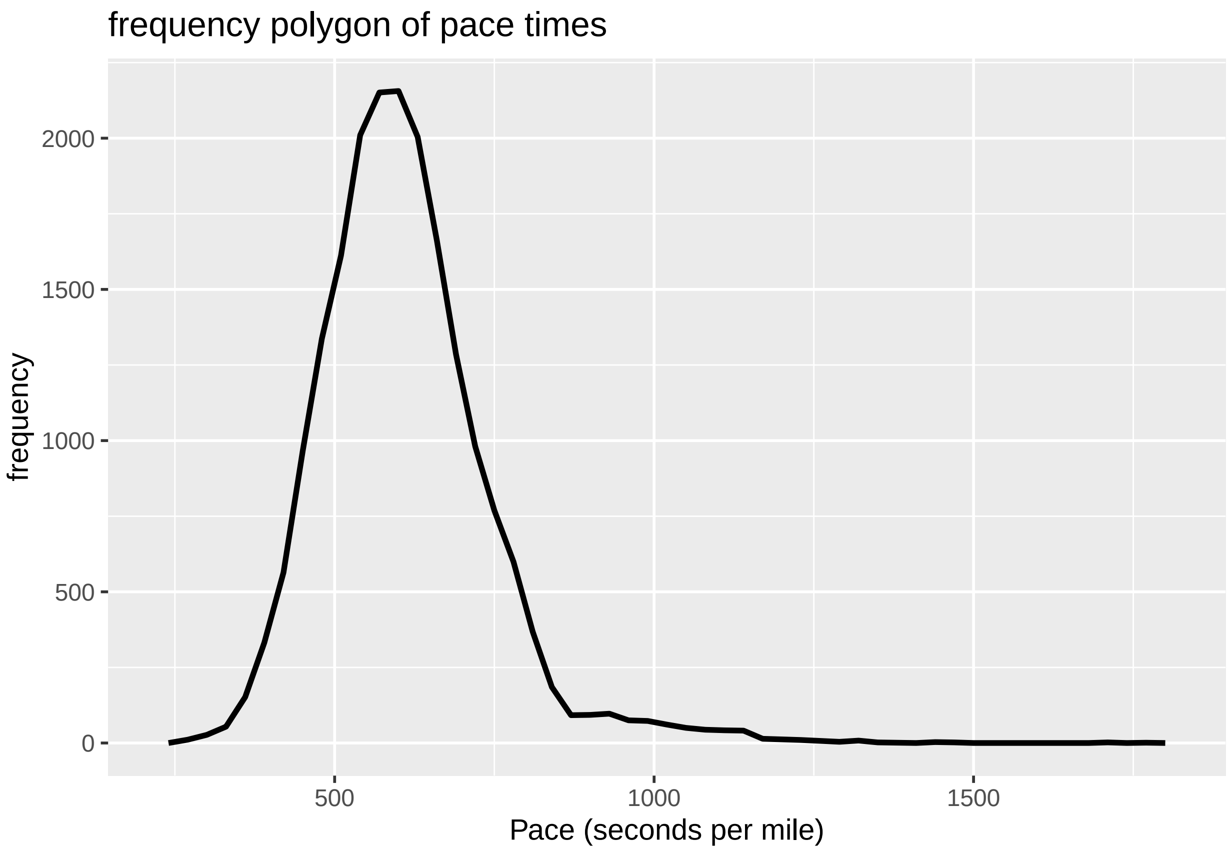



```r
# histogram
ggplot(data = run17, aes(x = pace_sec)) +
  geom_histogram(binwidth = 20) +
  labs(title = "histogram of pace times", x = "pace (seconds per mile)", y = "frequency")
```




```r
# frequency Polygon
ggplot(run17, aes(x = pace_sec)) +
  geom_freqpoly(binwidth = 30, size = 1) +
  labs(title = "frequency polygon of pace times", x = "Pace (seconds per mile)", y = "frequency")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Task 8: Have R compute the mean, median, and standard deviation of the pace-time among all runners in the Cherry Blossom Run. Also have R compute the 0.025, the 0.25, the 0.75, and the 0.975 quantiles for the same variable.

(See the R Code Cheat Sheet under the R Help Documents tab in Blackboard.)

```
#mean, median, sd
mean_pace <- mean(run17$pace_sec, na.rm = TRUE)
median_pace <- median(run17$pace_sec, na.rm = TRUE)
sd_pace <- sd(run17$pace_sec, na.rm = TRUE)

mean_pace
```

```
## [1] 608.688
```

```
median_pace
```

```
## [1] 596
```

```
sd_pace
```

```
## [1] 129.9132
```

```
#quantile ranges
quantile_results <- run17 %>%
  summarize(
    q025 = quantile(pace_sec, 0.025, na.rm = TRUE),
    q25 = quantile(pace_sec, 0.25, na.rm = TRUE),
    q75 = quantile(pace_sec, 0.75, na.rm = TRUE),
    q975 = quantile(pace_sec, 0.975, na.rm = TRUE)
```

```
  )

quantile_results
```

```
##   q025 q25 q75   q975
## 1  399 524 674 931.15
```

Task 9: Write a description of the distribution of pace-times for all runners in the race.

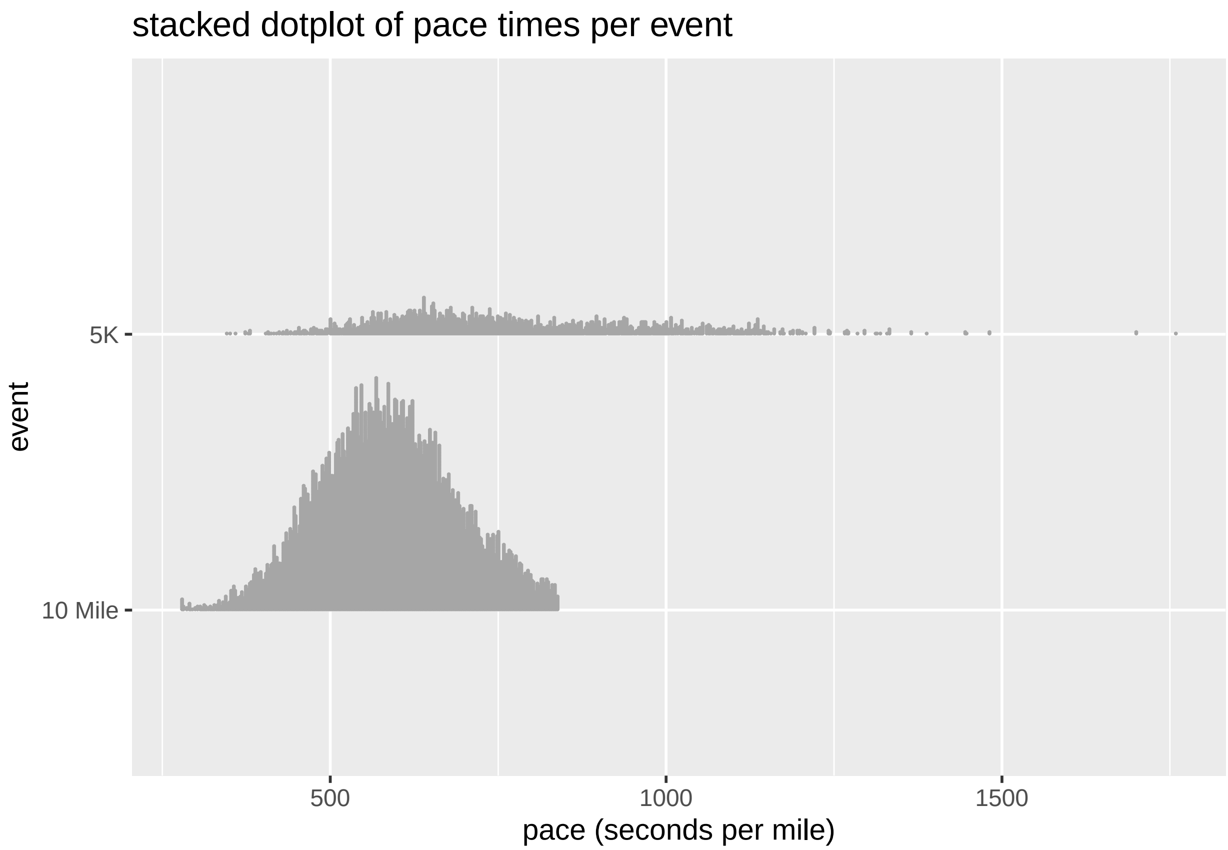
• Model your description on the one we did in class for ages of Oscar winners,linked below. • Make sure that you are addressing the key features of a numerical distribution – a w/w/w/w sentence, shape, center, spread, outliers (if any), other interesting features. • Write in complete sentences and use the context of the situation. (i.e. speak about runners and their pace times, not just about the dots in the dotplot, etc.)

Useful resources: Check Blackboard for our descriptions of single quantitative variables, as well as guidelines for how to write these descriptions.

The frequency polygon of pace times for the 2017 cherry blossom run shows a right skewed distribution, with most runners completing their miles around 500-600 seconds per mile. The peak frequency occurs just above 2000 runners, indicating this is the most common pace range. As pace time increases beyond 600 seconds, the frequency sharply declines afterwards.

Task 10: Have R make a stacked dotplot of pace time for each event. To make the graph, use `ggplot` with `pace_sec` as the x-variable and `event` as the y-variable. Use the `geom_dots()` geom.

```
# is this right?
ggplot(data = run17, aes(x = pace_sec, y = event)) +
  geom_dots() +
  labs(title = "stacked dotplot of pace times per event", x = "pace (seconds per mile)", y = "event")
```



Task 11: Next, have R compute the mean, median, standard deviation of pace time by event. Also compute the 0.025m the 0.25, the 0.75, and the 0.975 quantiles. To calculate the statistics, see the R cheat-sheet for how to compute statistics grouped by a categorical variable.

```
#done
#mean,median,sd for pace time by event
res <- run17 %>%
  group_by(event) %>%
  summarize(
    mean_PTE = mean(pace_sec, na.rm = TRUE),
    median_PTE = median(pace_sec, na.rm = TRUE),
    sd_PTE = sd(pace_sec, na.rm = TRUE)
  )

res
```

```
## # A tibble: 2 x 4
##   event   mean_PTE median_PTE sd_PTE
##   <chr>      <dbl>      <dbl>  <dbl>
## 1 10 Mile     586.        584   101.
## 2 5K          763.        725   189.
```

```
#quantiles ----------------->
PT_quantile <- run17 %>%
  group_by(event) %>%
  summarize(
    q025 = quantile(pace_sec, 0.025, na.rm = TRUE),
```

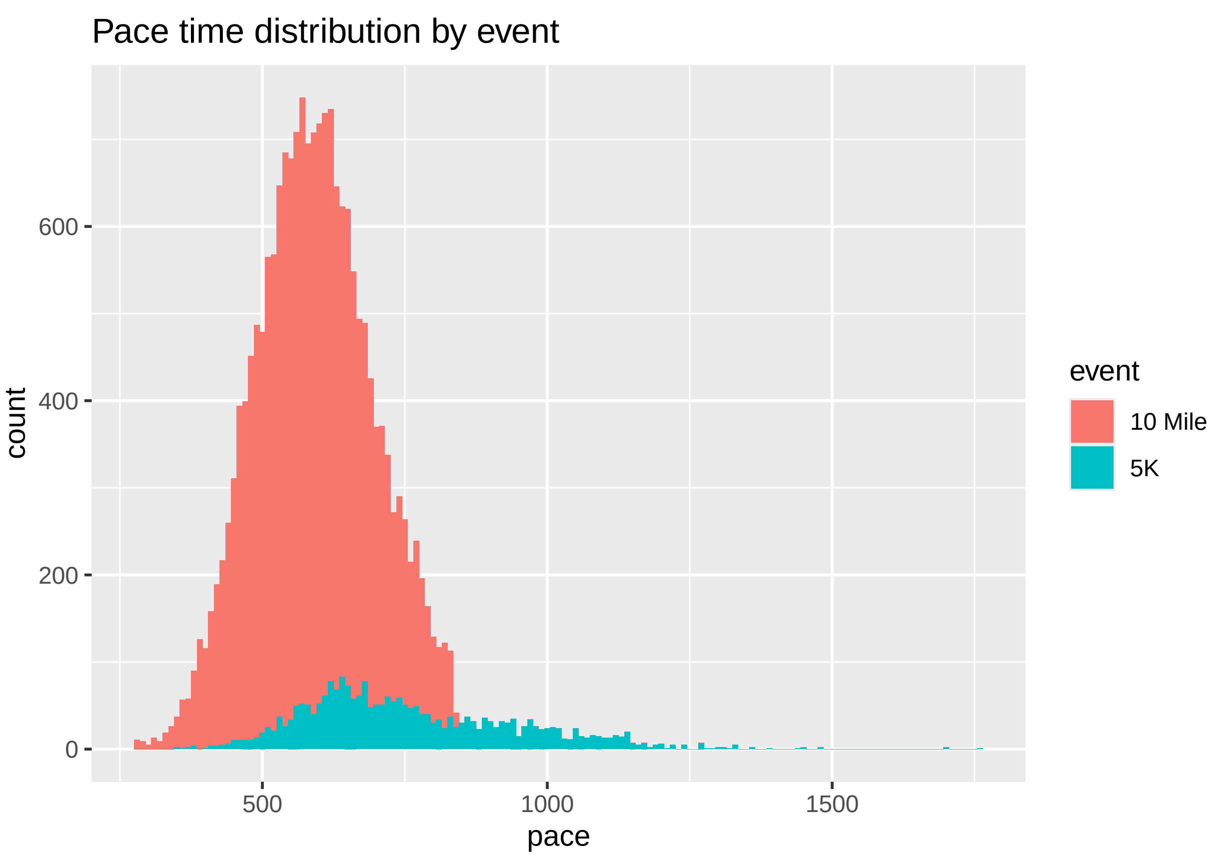
```
    q25 = quantile(pace_sec, 0.25, na.rm = TRUE),
    q75 = quantile(pace_sec, 0.75, na.rm = TRUE),
    q975 = quantile(pace_sec, 0.975, na.rm = TRUE)
  )
PT_quantile
```

```
## # A tibble: 2 x 5
##   event    q025   q25   q75  q975
##   <chr>   <dbl> <dbl> <dbl> <dbl>
## 1 10 Mile   394   516   654   790
## 2 5K        479   625   889  1161
```

Task 12: Write a paragraph where you *compare* the pace-times of 10 Mile runners with the pace-times of the 5K runners. You should compare the distributions using the key features of center, shape, and spread. Begin with a who/what/where sentence. After that, highlight the KEY DIFFERENCE OR SIMILARITY you notice, backing up what you say with the statistics you have calculated above. Then highlight what you think is the next most important thing to point out, again backing up what you say, and so on. . .

made a different graph below for better visual (also wanted to try using some graphs for more practicce)

```
ggplot(run17, aes(x= pace_sec, fill = event)) + geom_histogram(binwidth = 10) + labs(title = "Pace time
```



The distribution shows the 5k race is slightly right skewed and more spread out where the 10k race is bell shaped, with a large concentration of runners maintaining an average pace around 600 seconds per mile. However the 10k race had significantly more participants than the 5K race, which can be seen in the higher density of runners in that axis (count). While both distributions show most runners finishing around 600 seconds per mile, the spread differs between the two events. The 5k race has a wider spread indicating more variation in times among participants. where as, the 10k race maintains a more consistent distribution from beginning to end, leveling off as pace increases.